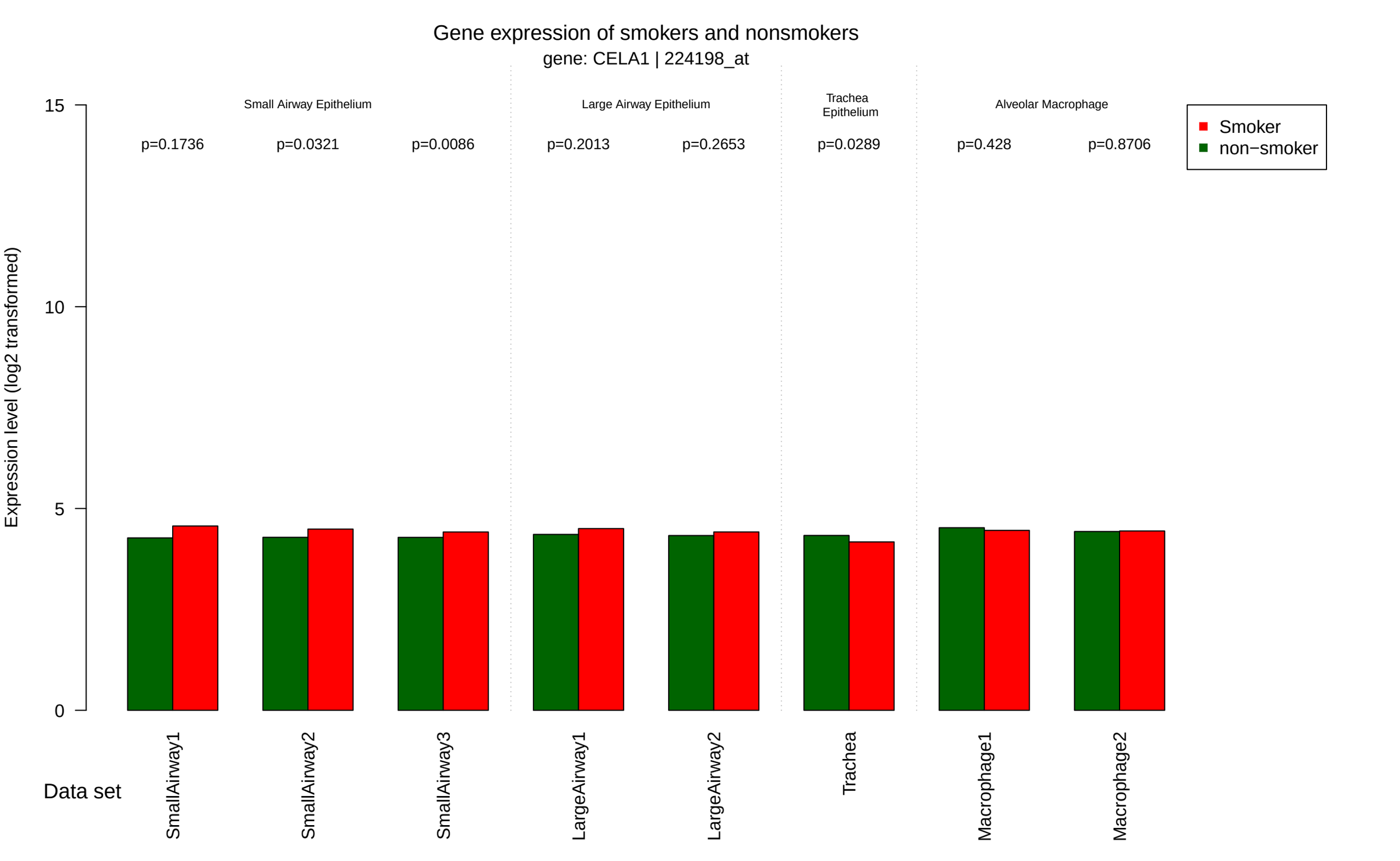
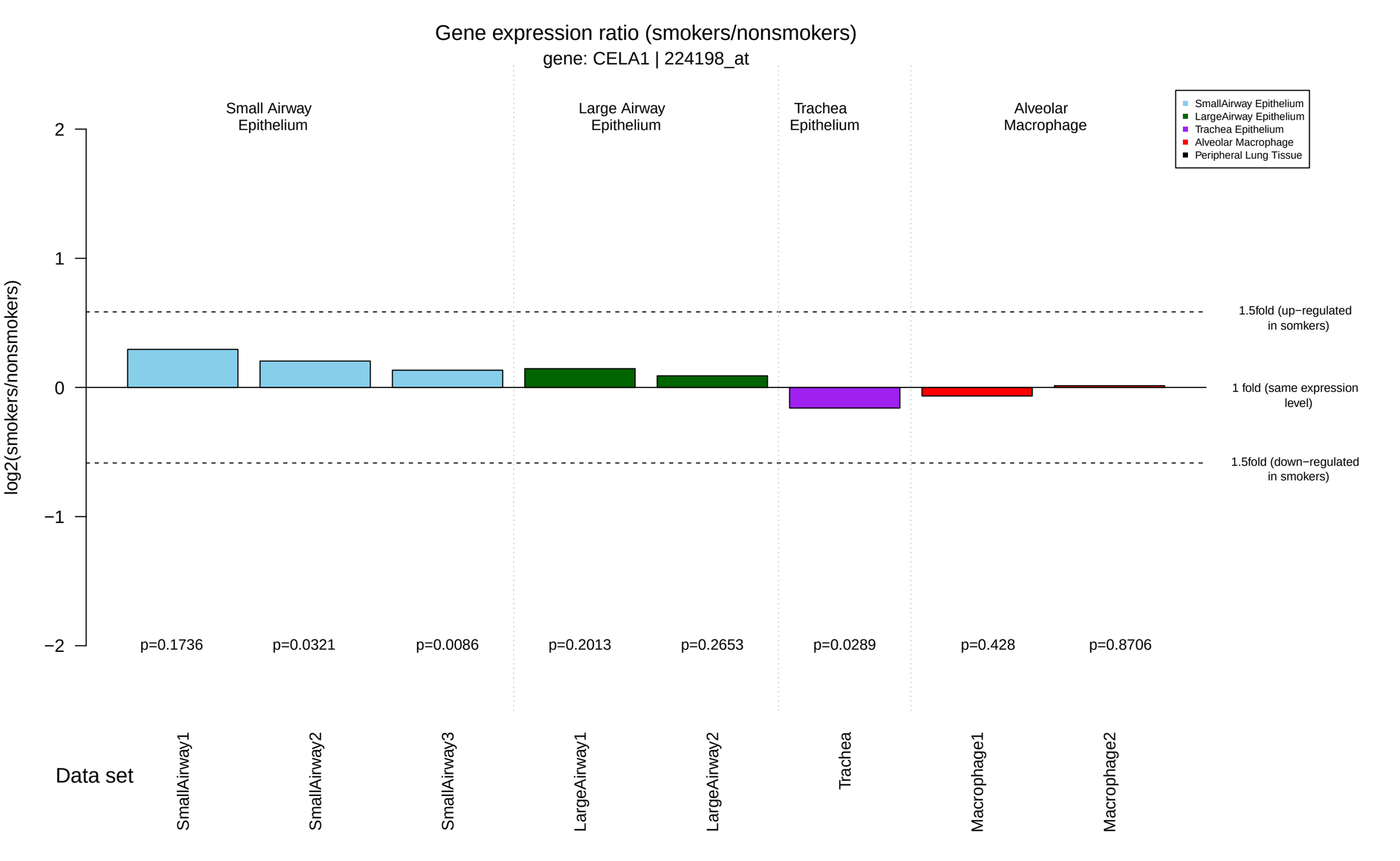
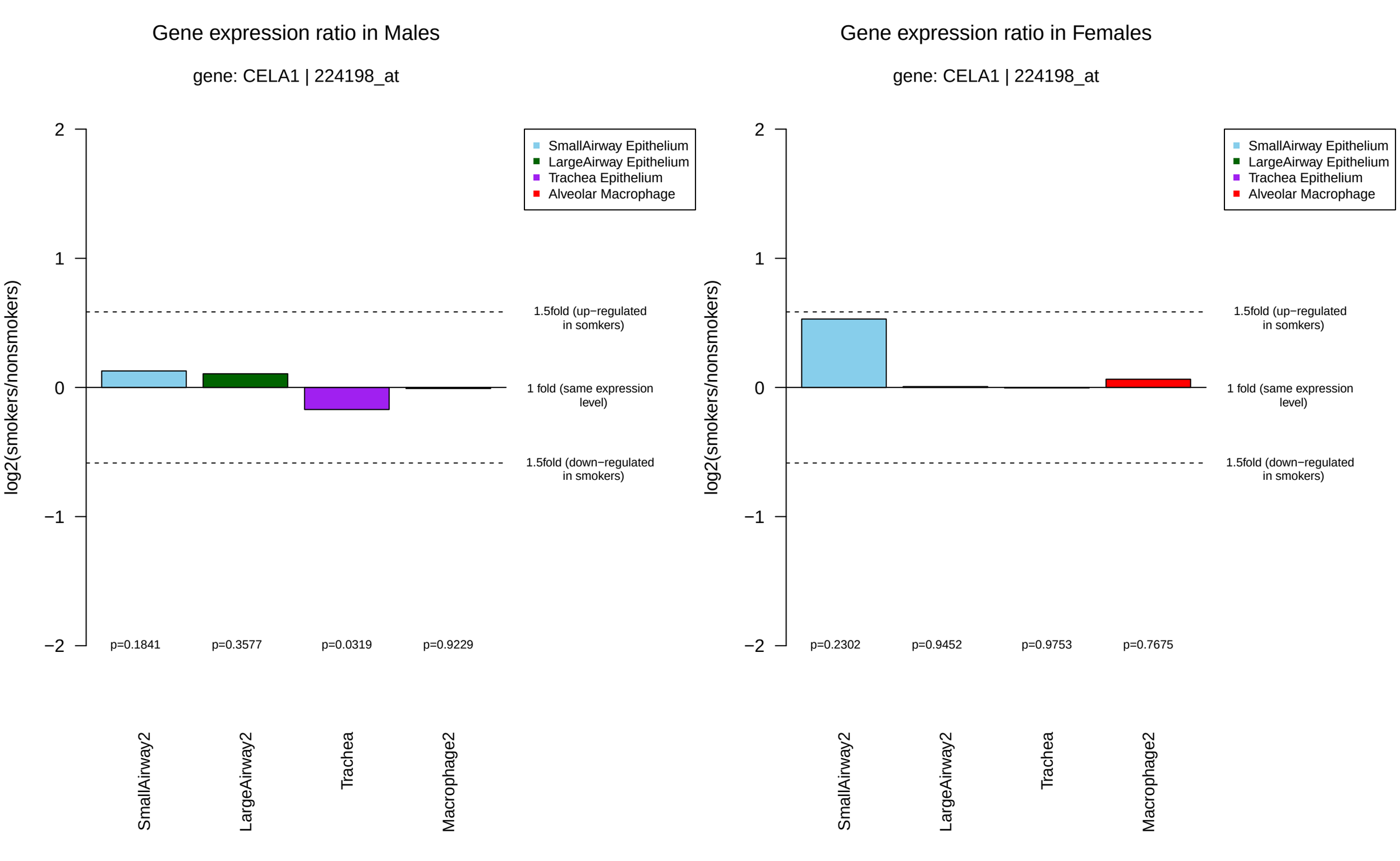
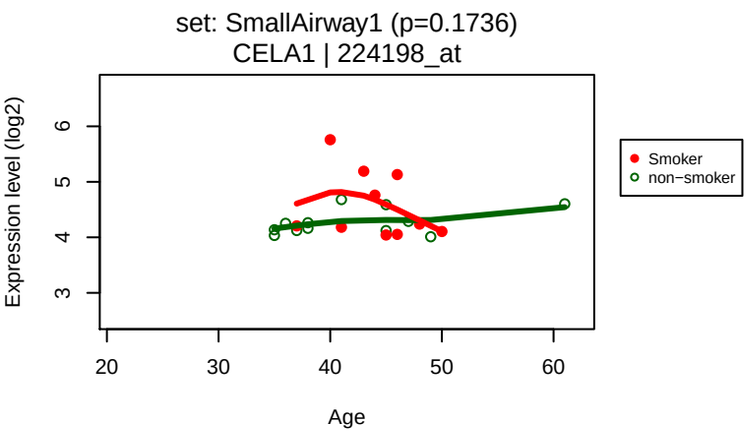
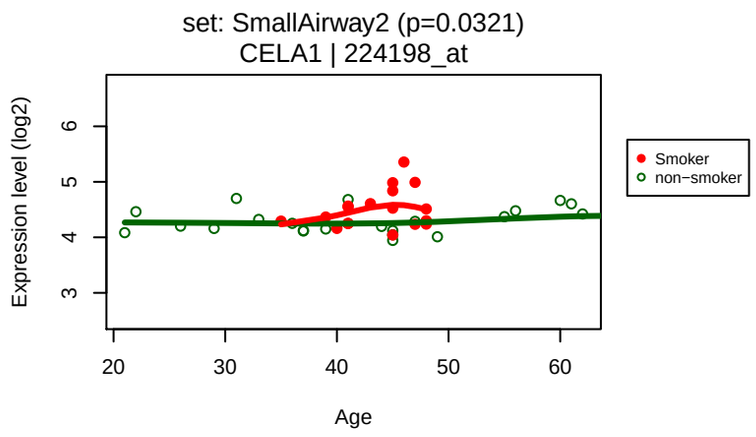
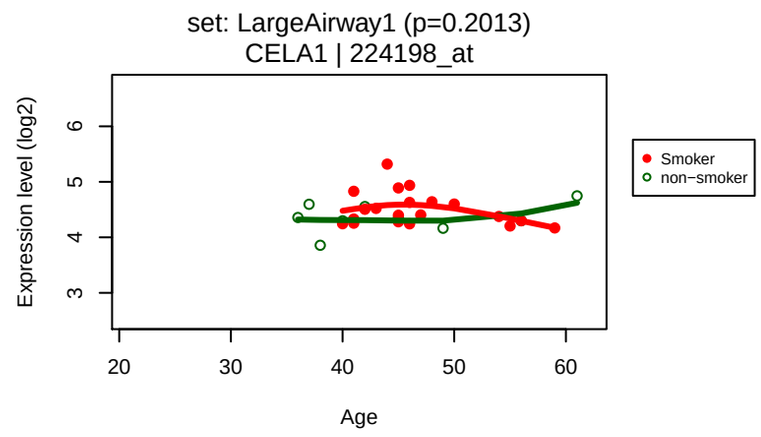

Gene expression of smokers and nonsmokers
gene: CELA1 | 224198_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1736
p=0.0321
p=0.0086
p=0.2013
p=0.2653
p=0.0289
p=0.428
p=0.8706
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

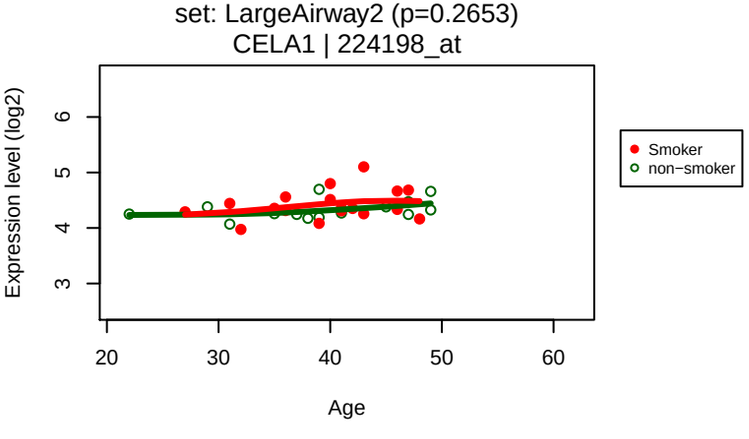

Gene expression ratio (smokers/nonsmokers)
gene: CELA1 | 224198_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1736
p=0.0321
p=0.0086
p=0.2013
p=0.2653
p=0.0289
p=0.428
p=0.8706
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

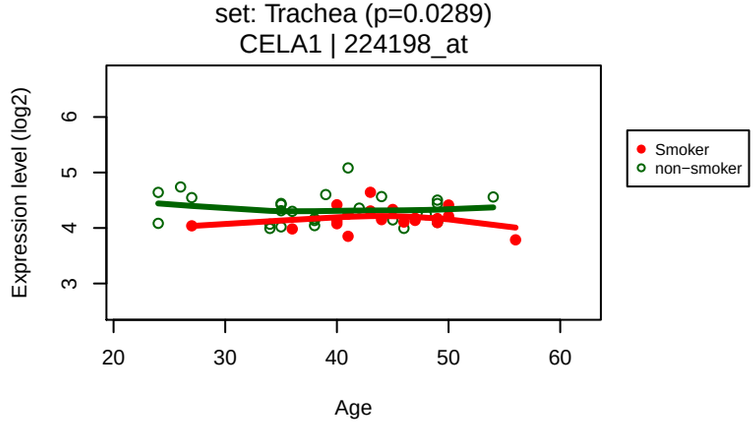

Gene expression ratio in Males
gene: CELA1 | 224198_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1841
p=0.3577
p=0.0319
p=0.9229
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CELA1 | 224198_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2302
p=0.9452
p=0.9753
p=0.7675
SmallAirway2
LargeAirway2
Trachea
Macrophage2

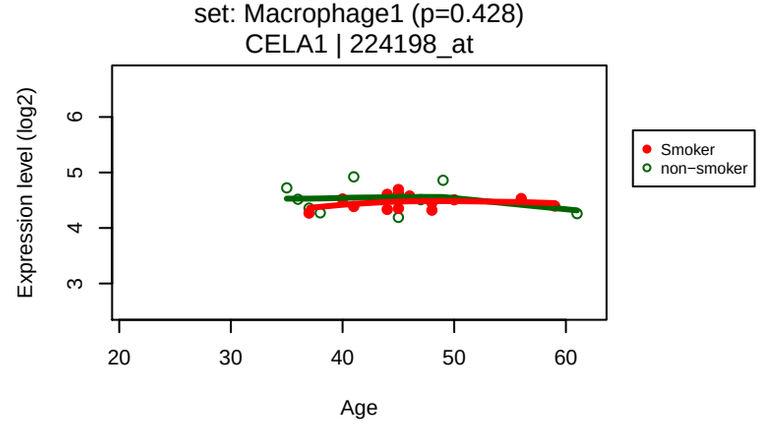

set: SmallAirway1 (p=0.1736)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker

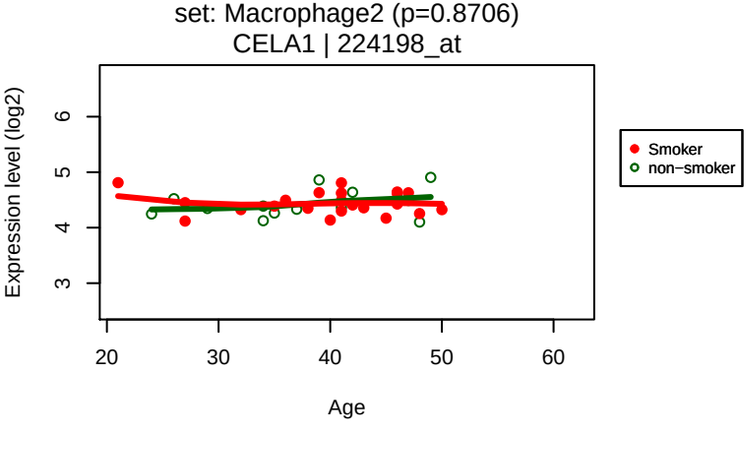

set: SmallAirway2 (p=0.0321)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker

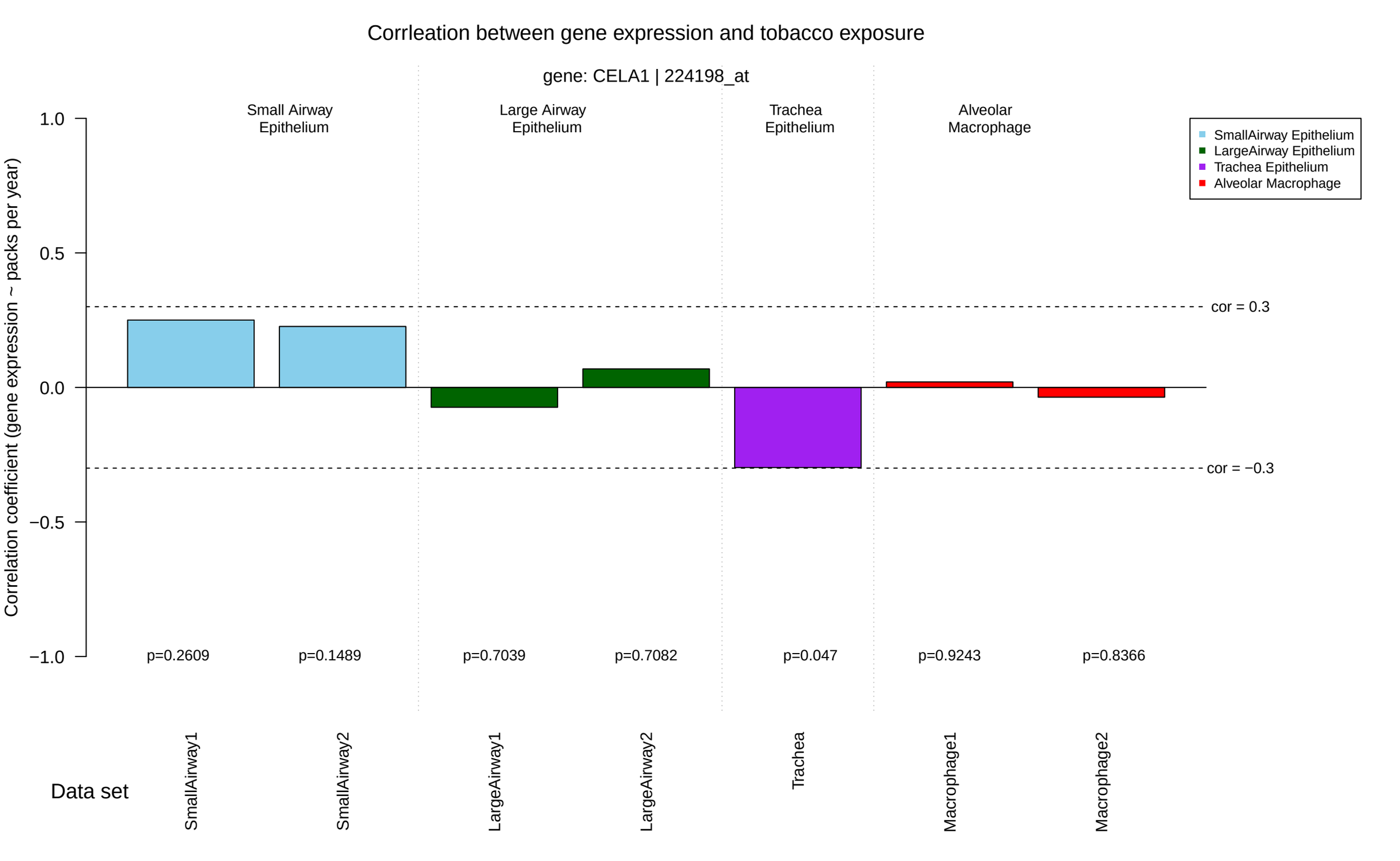

set: LargeAirway1 (p=0.2013)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2653)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0289)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.428)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8706)
CELA1 | 224198_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CELA1 | 224198_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2609
p=0.1489
p=0.7039
p=0.7082
p=0.047
p=0.9243
p=0.8366
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2